Nguyen Manh Hiep

Software Engineer

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SUMMARY

AI-native full-stack software engineer: I bring LLMs and automation into products, and use AI agents as everyday development tools.
Short term (1–2 years): Join a product team to build and run production AI features (LLM integration, agents, RAG) and the backend behind them, while raising my own standards in testing, code review and system monitoring.
Long term (3–5 years): Grow into a Senior Software Engineer specialising in AI systems, owning a system's technical direction from design to operation and helping the team use AI safely in its development workflow.

EXPERIENCE

ChimAI | Notarex - notarial document digitization platform 03/2026 - 09/2026

Software Engineer, ChimAI JSC (Full-stack + AI document processing + Infrastructure):

- **Description:** A platform that digitizes notarial paperwork for notary offices in Vietnam, covering configurable form templates, AI-driven OCR extraction and a full document edit, save and download cycle.
- **Technologies & Tools:** Next.js 14, TypeScript, FastAPI, SQLAlchemy async, PostgreSQL 16, Celery, Redis, MinIO, OnlyOffice
- **Key Responsibilities & Contributions:**
  - **AI OCR Pipeline:** Generated each document type's extraction prompt from its template configuration, so new document types are added through configuration alone; national IDs are read from their QR code and skip the model.
  - **Document Workflow & Authorization:** Built the flow from uploaded photos through human review to a JWT-protected OnlyOffice edit-save-download cycle with docxtpl DOCX generation, under a three-role model scoped per notary office.
  - **Background Processing:** Built a Celery and Redis task queue on a ZSET pending queue with an atomic active-counter, plus heartbeat reconciliation that detects and reruns zombie jobs so no record gets stuck mid-OCR.
  - **Requirements & Delivery:** Collected requirements directly with notary offices, scoped each release in sprints in a 2 to 3 person team with no dedicated PM, and shipped on Docker behind Nginx with MinIO storage and GitHub Actions CI/CD.

PROJECTS

SmartSales - sales platform for small online shops 2026

Solo developer (Architecture + Backend + AI integration):

- **Description:** Sales software for small Vietnamese shops that puts Zalo chat, orders, calls and livestream selling in one place. I built it alone, from the architecture to the UI and the AI.
- **Technologies & Tools:** Node.js 22, Express, PostgreSQL 16 + pgvector, Redis, Socket.IO, zca-js, JsSIP (SIP over WebRTC), Docker Compose, PM2
- **Key Responsibilities & Contributions:**
  - **Modular Architecture:** Split the system into 7 staff modules on one PostgreSQL database. Each module writes only to its own schema but can join across the others, so one customer profile pulls in orders, chats and calls; all modules share one sign-in and one UI frame.
  - **AI Gateway & RAG:** Routed every AI call through one gateway that picks a model per task from .env (Gemini, OpenAI, Groq) and falls back to the next provider on failure. Documents and products sit in a RAG index searched by vectors and keywords together, which feeds reply suggestions and the shop chatbot.
  - **Zalo & Call Integration:** Wrote a CLI on the open-source zca-js to connect personal Zalo accounts, replacing a licensed tool. Calls are recorded on two channels so the transcript knows who said what, and the AI scores staff attitude on 7 criteria.
  - **Security & Testing:** Kept 76 test files and an end-to-end smoke test that rerun on every change, fixed all 61 issues found in two security reviews, and put session, CSRF and Socket.IO checks in one shared layer that every module mounts.

SenAI - real-time multilingual meeting translation 2026 - present

Frontend, Backend and AI Integration Owner (Real-time systems + LLM orchestration + Infrastructure):

- **Description:** Live meeting translation by a 5-person team for VAIC 2026 (three awards), running at app.senai.online.
- **Technologies & Tools:** Next.js 16, TypeScript, FastAPI, PostgreSQL + pgvector, LiveKit, vLLM, Soniox, ElevenLabs
- **Key Responsibilities & Contributions:**
  - **AI/LLM Integration:** Built a model router for ASR, translation, TTS and LLM that switches provider at runtime between self-hosted vLLM and commercial APIs, falling back to the self-hosted model when a key is missing or a provider fails.
  - **Translation Pipeline:** Ran an independent ASR and translation pipeline per microphone to separate speakers, pairing fast translation with LLM refinement against a contextual glossary.
  - **Meeting Intelligence:** Added Soniox transcription that separates three or more speakers, LLM-written minutes with an assistant that answers questions about them, and per-speaker voiced translation with ElevenLabs.
  - **Delivery & Security:** Deployed through GitHub Actions over SSH with a forced command that skips API restarts for doc-only commits, revoked refresh tokens on logout, and kept 37 UAT suites with 147 real, non-mocked tests.

Ely - personal AI assistant built on OpenClaw 2026

Extended the open-source OpenClaw agent with Telegram commands, CDP browser automation, cron jobs for daily chores, my own skills that dispatch and supervise Claude Code tasks, and a self-restarting gateway.

EDUCATION

Hanoi University of Science and Technology (HUST) 2021 - 2026

Computer Science

HONORS & AWARDS

Third Prize (top 3 overall of 300+ teams), Vietnam AI Innovation Challenge 2026

Vietnam AI Stars Award, Vietnam AI Innovation Challenge 2026

AI Singapore Award, Vietnam AI Innovation Challenge 2026

TECHNICAL SKILLS

AI-assisted development	Claude Code (agents, skills, hooks), OpenClaw agents, MCP servers, Cursor, Copilot, prompt & context engineering, RAG, LLM orchestration, model routing with fallback, vLLM
Frontend	React 19/18, Next.js 16/14, TypeScript, Vite, Tailwind v4, Zustand, PWA / offline-first
Backend	FastAPI, Python, Frappe v15, REST API design, Node.js / Express / NestJS, Spring Boot, WebSocket / Socket.IO, LiveKit, Celery, Redis queues, JWT / OAuth, RBAC, webhooks
Databases	PostgreSQL (+ pgvector), MariaDB, MySQL, SQLite, SQLAlchemy 2.0 async, MinIO (S3)
DevOps & Cloud	Docker Compose, Nginx, systemd, TLS, VPS deployment, GitHub Actions CI/CD
Engineering practice	System design, SOLID, design patterns, code review, idempotency & concurrency control, workflow & scheduling engines, non-mocked integration testing, SRS / Use Case / ERD
Soft Skills	Critical thinking, technical research, cross-functional teamwork, fast adaptability